

FCU3

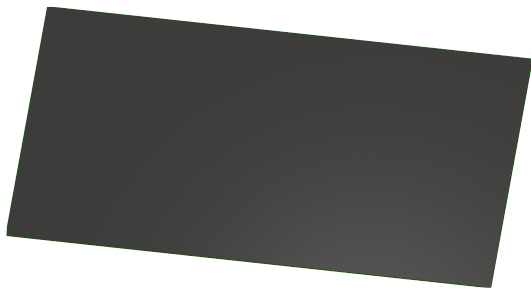
Variant: DRAFT

2025-11-24

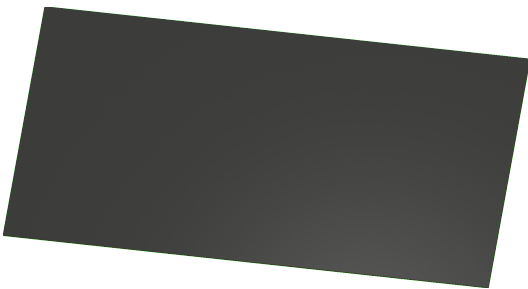
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.  
PRELIMINARY - Close to final schematic.  
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.  
RELEASED - A board with this schematic has been sent to production.

Date: 24-Nov-2025

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

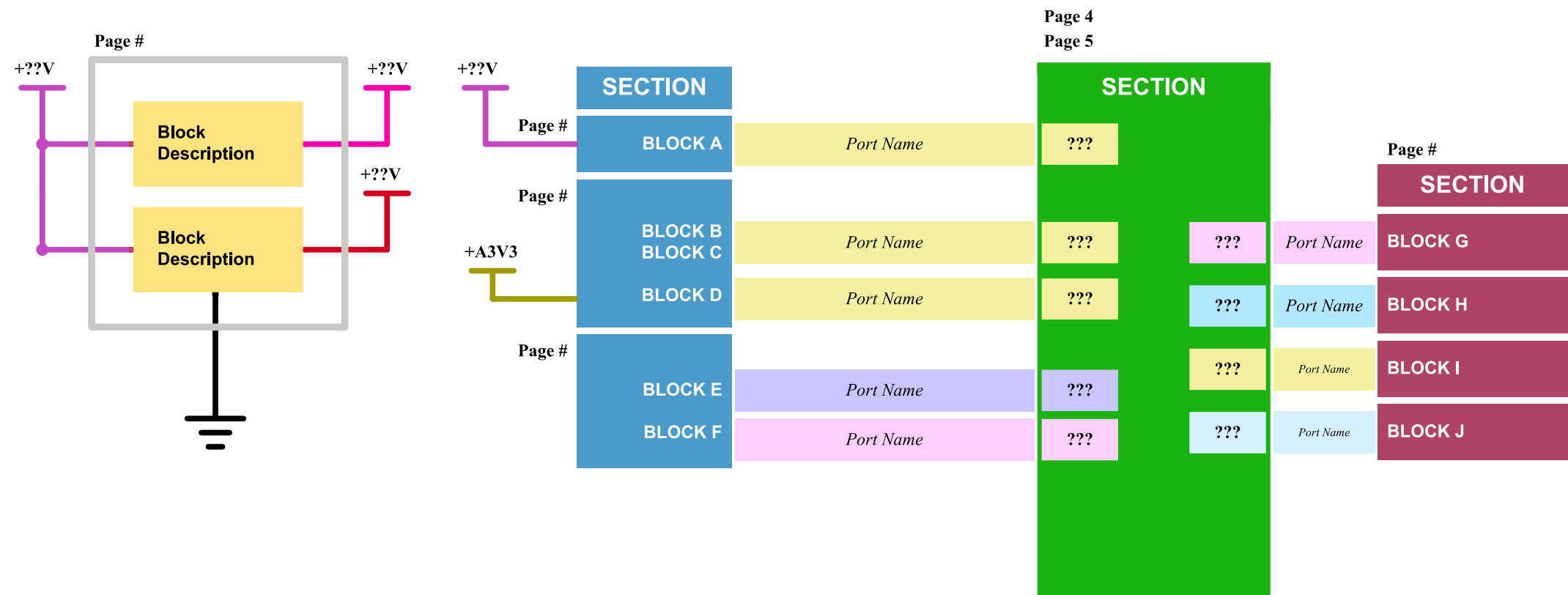
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

|  |                  |                                    |                         |                                      |                             |
|--|------------------|------------------------------------|-------------------------|--------------------------------------|-----------------------------|
|  | Comments:        | Company:<br>Kaldstrand Electronics |                         | Variant:<br>DRAFT                    | Git Hash:<br>70c4a9a        |
|  |                  | Board Name:<br>FCU3                |                         | Project Name:<br>Flight Controller 3 |                             |
|  | Sheet Title:     | File Name:<br>FCU3.kicad_sch       | Designer:<br>Kaldstrand | Date:<br>2025-01-12                  | Revision:<br>+ (Unreleased) |
|  | Sheet Path:<br>/ |                                    | Reviewer:               | Size:<br>A3                          | Sheet:<br>1 of 7            |

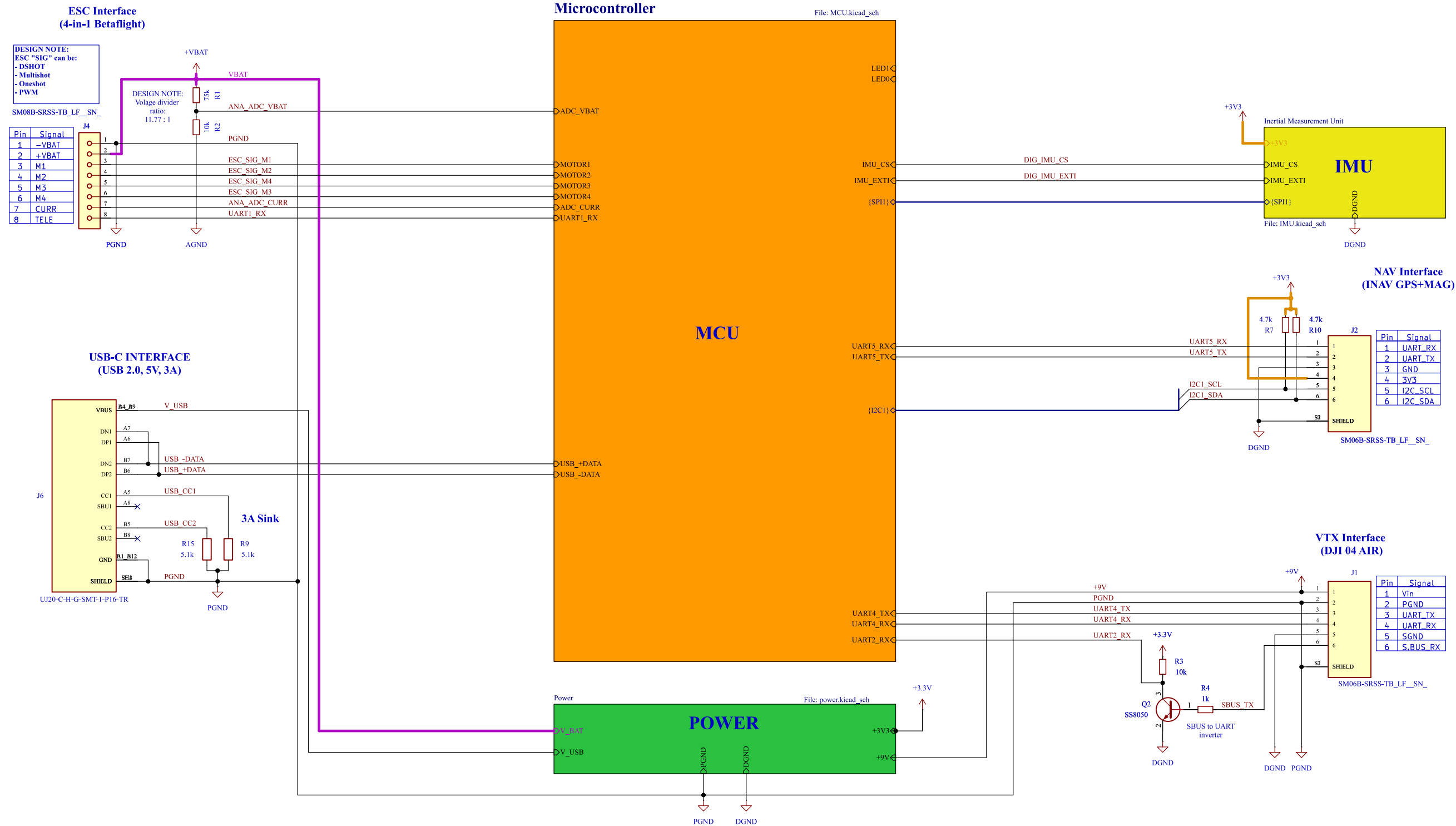
# [2] Block Diagram



| Target specifications: |           |
|------------------------|-----------|
| Input voltage:         | ?? - ?? V |
| Spec 2                 | ??        |
| Spec 3                 | ??        |
| Spec 4                 | ??        |

|  |                                |                                       |                                    |                     |                                      |                      |
|--|--------------------------------|---------------------------------------|------------------------------------|---------------------|--------------------------------------|----------------------|
|  | Comments:                      |                                       | Company:<br>Kaldstrand Electronics |                     | Variant:<br>DRAFT                    | Git Hash:<br>70c4a9a |
|  |                                |                                       | Board Name:<br>FCU3                |                     | Project Name:<br>Flight Controller 3 |                      |
|  | Sheet Title:<br>Block Diagram  | File Name:<br>Block Diagram.kicad_sch | Designer:<br>Kaldstrand            | Date:<br>2025-01-12 | Revision:<br>+ (Unreleased)          |                      |
|  | Sheet Path:<br>/Block Diagram/ |                                       | Reviewer:                          | Size:<br>A3         | Sheet:<br>2 of 7                     |                      |

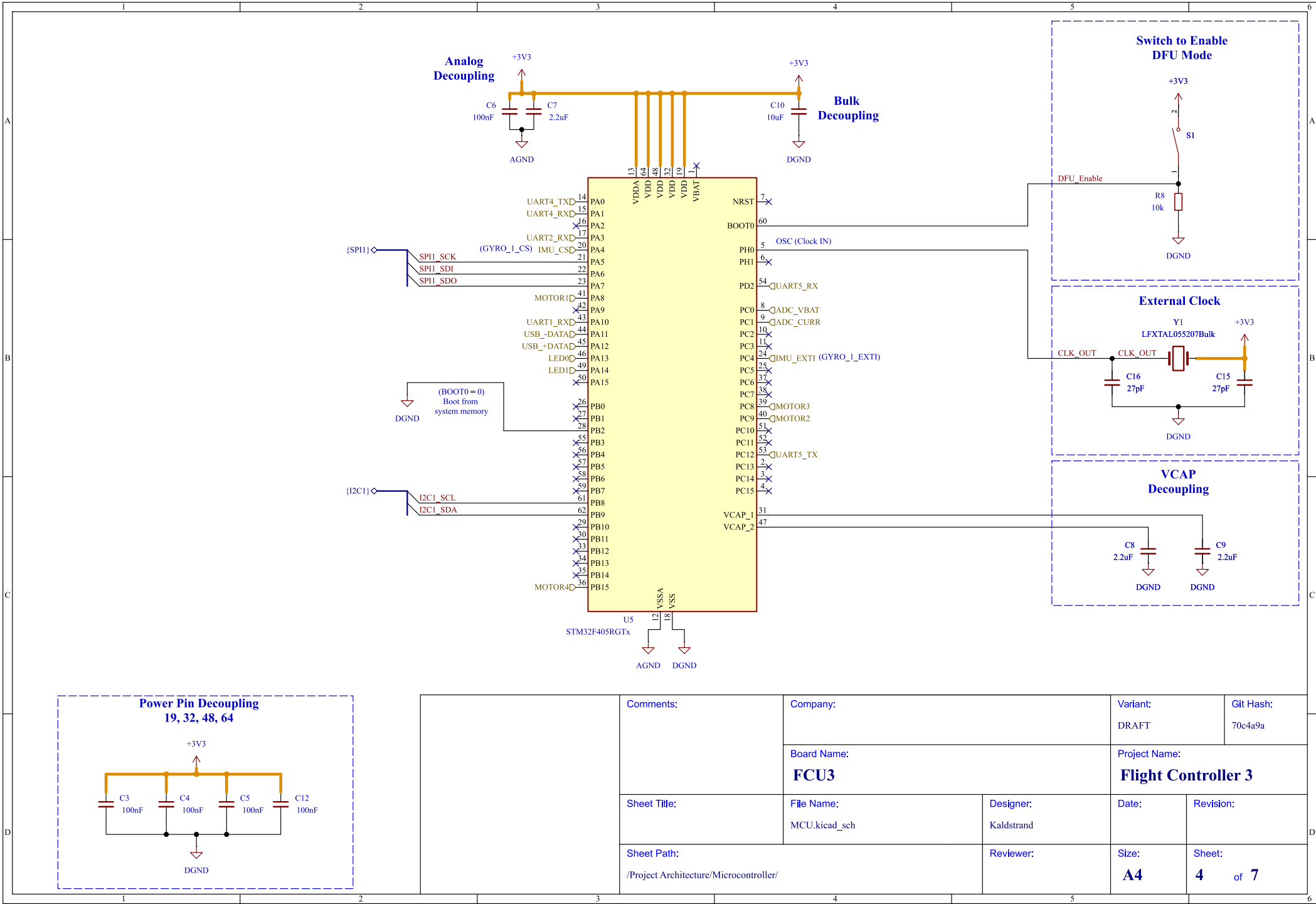
# [3] Project Architecture

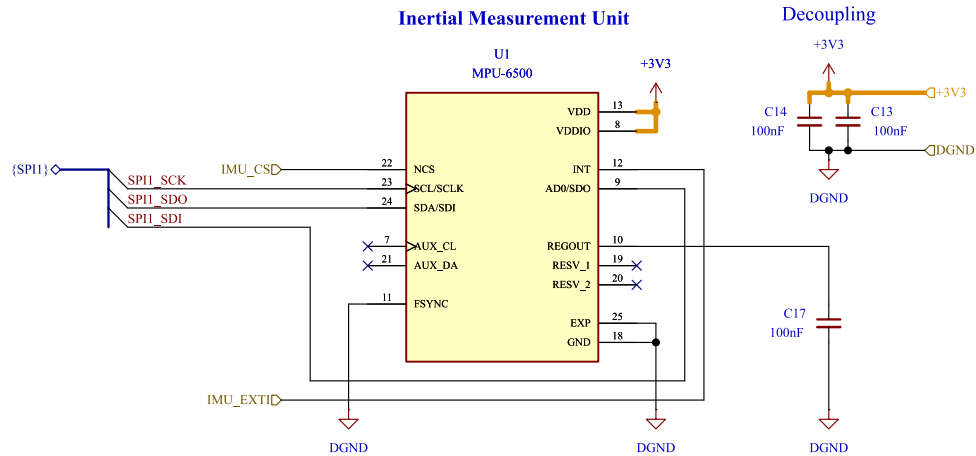


|                                       |                                           |                                              |                                             |                             |
|---------------------------------------|-------------------------------------------|----------------------------------------------|---------------------------------------------|-----------------------------|
| Comments:                             | Company:<br><b>Kaldstrand Electronics</b> |                                              | Variant:<br>DRAFT                           | Git Hash:<br>70c4a9a        |
|                                       | Board Name:<br><b>FCU3</b>                |                                              | Project Name:<br><b>Flight Controller 3</b> |                             |
|                                       | Sheet Title:<br>Project Architecture      | File Name:<br>Project Architecture.kicad_sch | Designer:<br>Kaldstrand                     | Date:<br>2025-01-12         |
| Sheet Path:<br>/Project Architecture/ |                                           | Reviewer:                                    | Size:<br><b>A3</b>                          | Revision:<br>+ (Unreleased) |
|                                       |                                           |                                              | Sheet:<br><b>3</b> of <b>7</b>              |                             |

# [9] Revision History

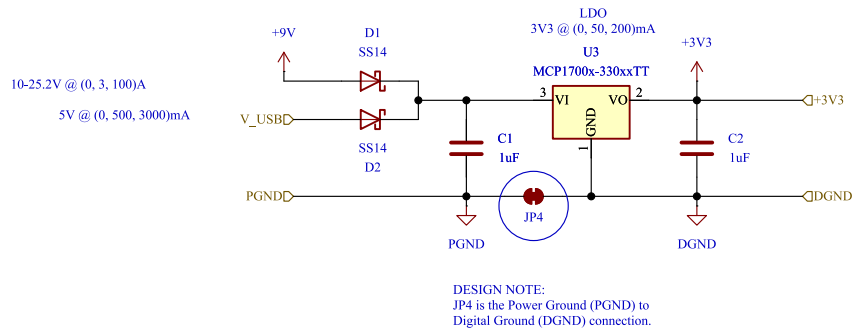
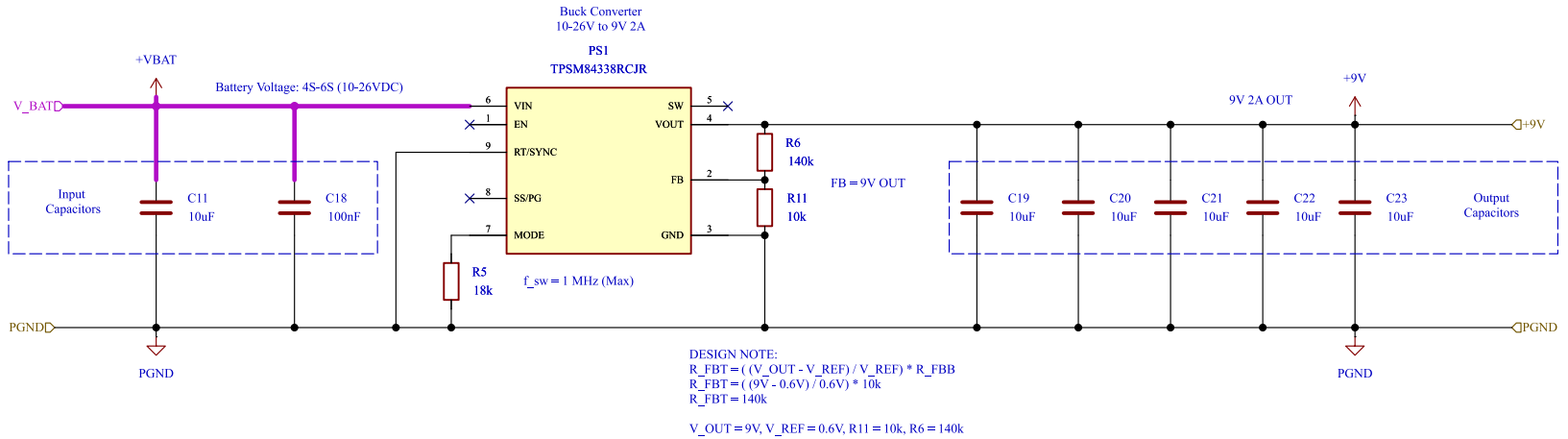
|  |                                   |                                          |                         |                                      |                             |
|--|-----------------------------------|------------------------------------------|-------------------------|--------------------------------------|-----------------------------|
|  | Comments:                         | Company:<br>Kaldstrand Electronics       |                         | Variant:<br>DRAFT                    | Git Hash:<br>70c4a9a        |
|  |                                   | Board Name:<br>FCU3                      |                         | Project Name:<br>Flight Controller 3 |                             |
|  | Sheet Title:<br>Revision History  | File Name:<br>Revision History.kicad_sch | Designer:<br>Kaldstrand | Date:<br>2025-01-12                  | Revision:<br>+ (Unreleased) |
|  | Sheet Path:<br>/Revision History/ |                                          | Reviewer:               | Size:<br>A4                          | Sheet:<br>9 of 7            |





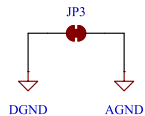
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|-----------------------------------------------------------------|--------------|-----------------------------|-------------------------|--------------------------------------|------------------|
|                                                                 | Comments:    | Company:                    |                         | Variant:                             | Git Hash:        |
|                                                                 |              |                             |                         | DRAFT                                | 70c4a9a          |
|                                                                 |              | Board Name:<br>FCU3         |                         | Project Name:<br>Flight Controller 3 |                  |
|                                                                 | Sheet Title: | File Name:<br>IMU.kicad_sch | Designer:<br>Kaldstrand | Date:                                | Revision:        |
| Sheet Path:<br>/Project Architecture/Inertial Measurement Unit/ |              | Reviewer:                   |                         | Size:<br>A4                          | Sheet:<br>5 of 7 |

To-Do:  
- Add all settings used in Vbat to 9V converter.  
- Add all settings used in 3.3V LDO.

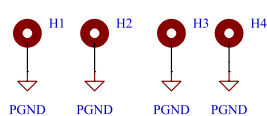


### Grounding Strategy

#### Ground Ties



#### Mounting Holes



Comments:

Company:

Variant:

Git Hash:

DRAFT

70c4a9a

Board Name:

FCU3

Project Name:

Flight Controller 3

Sheet Title:

File Name:

power.kicad\_sch

Designer:

Kaldstrand

Date:

Revision:

Sheet Path:

/Project Architecture/Power/

Reviewer:

Size:

A4

Sheet:

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